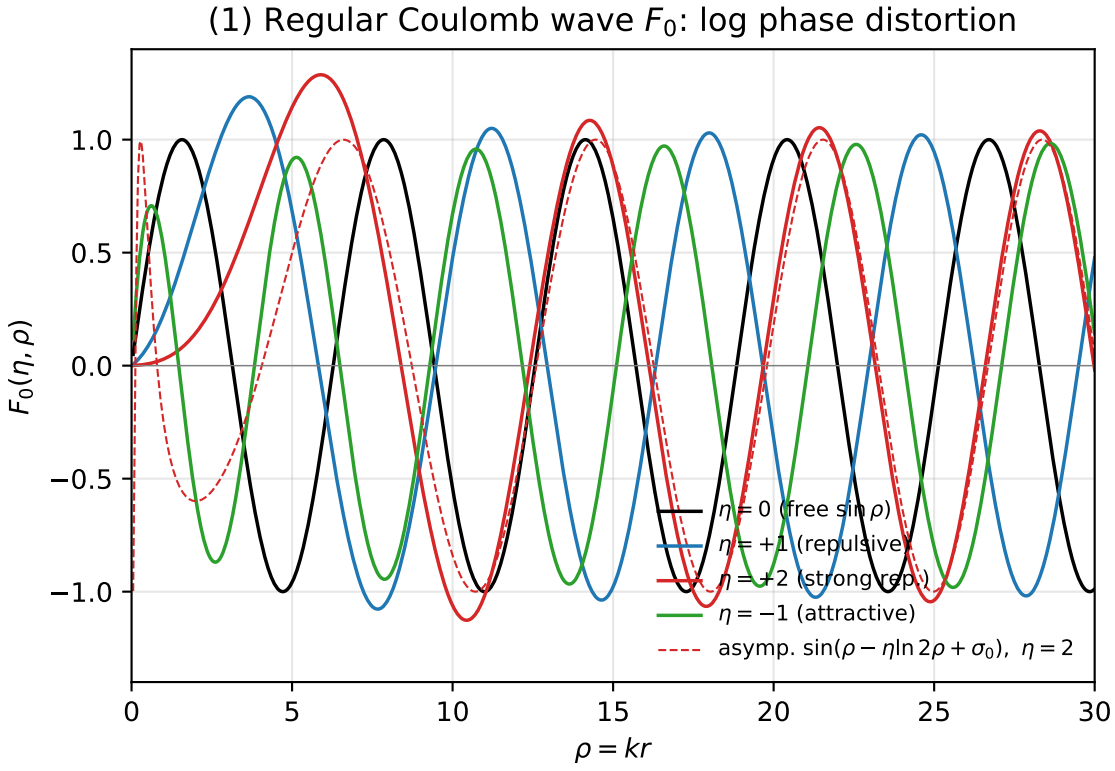
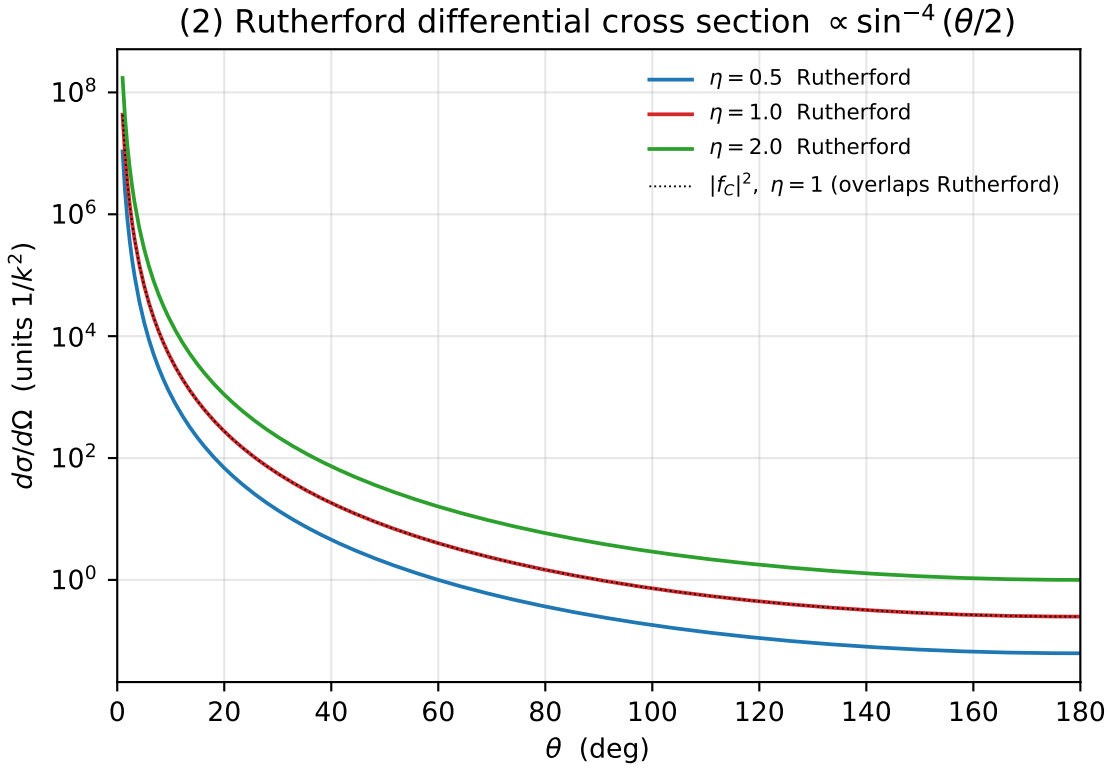
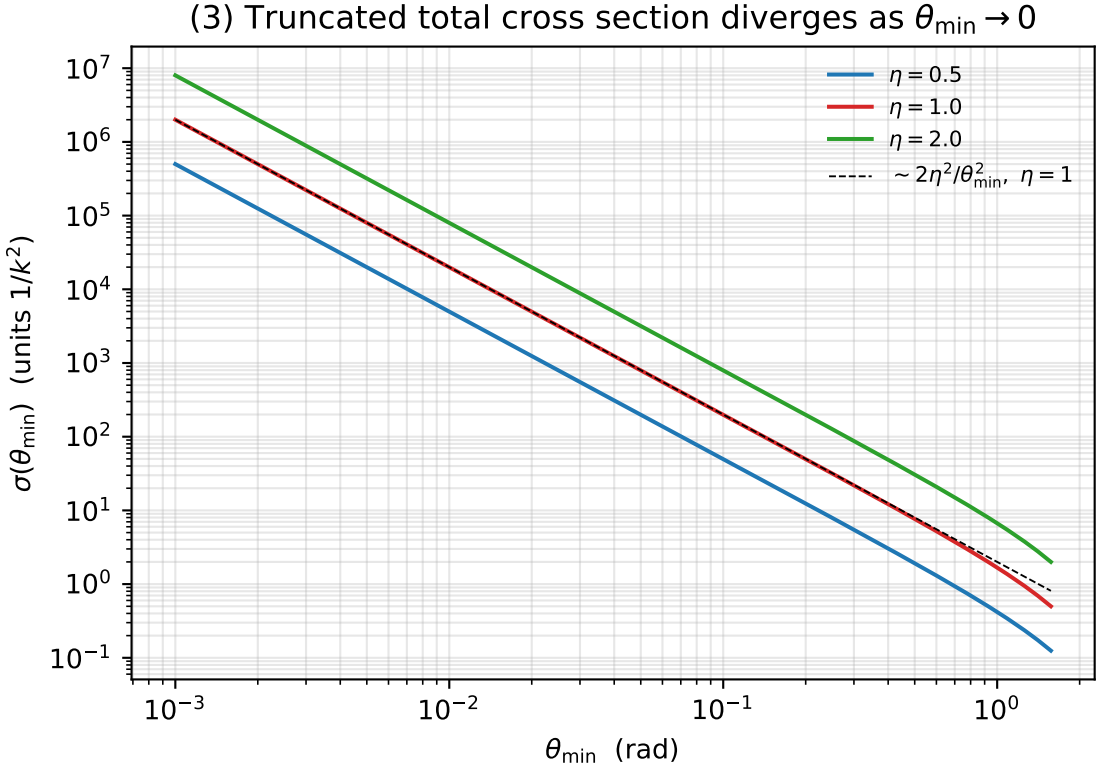
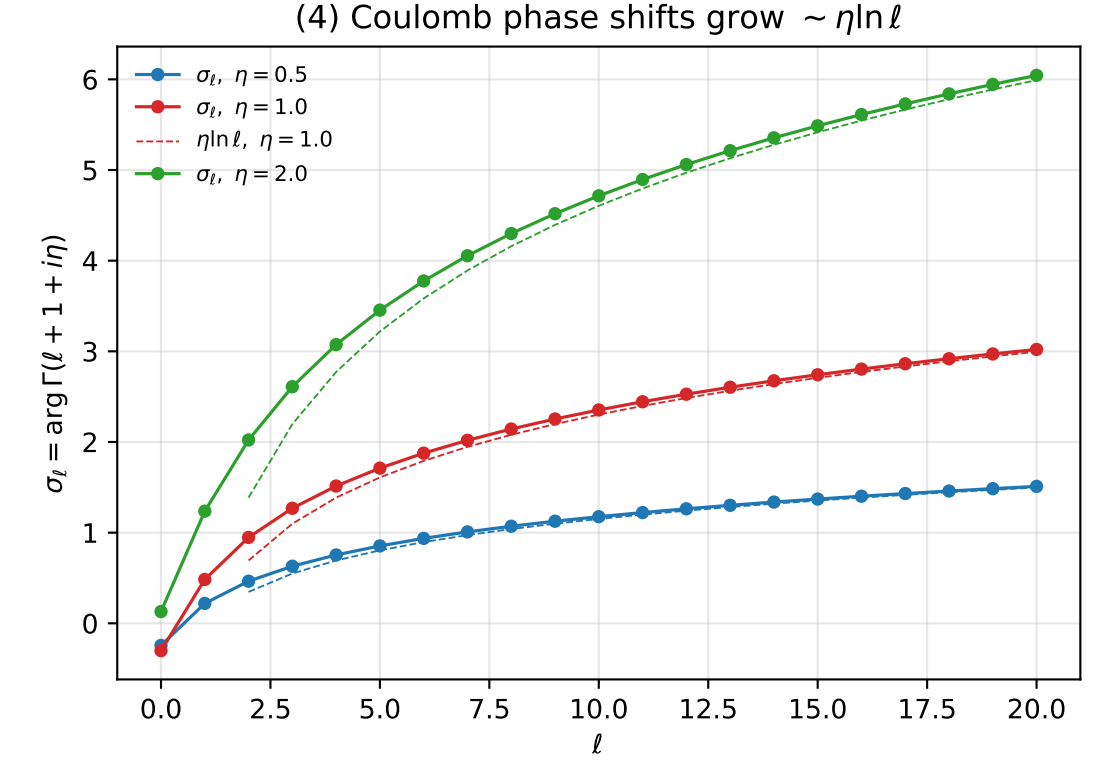

(1) Regular Coulomb wave F_0: log phase distortion
$F_0(\eta,\rho)$
$\rho = kr$
$\eta = 0$ (free $\sin\rho$)
$\eta = +1$ (repulsive)
$\eta = +2$ (strong rep.)
$\eta = -1$ (attractive)
asymp. $\sin(\rho - \eta\ln 2\rho + \sigma_0)$, $\eta = 2$


(2) Rutherford differential cross section $\propto \sin^{-4}(\theta/2)$
$d\sigma/d\Omega$ (units $1/k^2$)
θ (deg)
$\eta = 0.5$ Rutherford
$\eta = 1.0$ Rutherford
$\eta = 2.0$ Rutherford
$|f_C|^2$, $\eta = 1$ (overlaps Rutherford)


(3) Truncated total cross section diverges as $\theta_{\min} \to 0$
$\sigma(\theta_{\min})$ (units $1/k^2$)
$\theta_{\min}$ (rad)
$\eta = 0.5$
$\eta = 1.0$
$\eta = 2.0$
$\sim 2\eta^2/\theta_{\min}^2$, $\eta = 1$


(4) Coulomb phase shifts grow $\sim \eta\ln\ell$
$\sigma_\ell = \arg\Gamma(\ell + 1 + i\eta)$
ℓ
σ_ℓ, $\eta = 0.5$
σ_ℓ, $\eta = 1.0$
$\eta\ln\ell$, $\eta = 1.0$
σ_ℓ, $\eta = 2.0$